

Wavelet Analysis of GPS Data for Earthquake Prediction in Japan: Investigating Long-Term Patterns and Periodicity

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Abstract:

Earthquakes pose significant risks to human safety and infrastructure, making it crucial to understand and predict their occurrences and impacts. This is especially important in Japan, an extremely densely populated region, which is well known for its common high-magnitude earthquakes. This research aims to determine whether there is a long-term pattern in GPS data that can help predict whether an earthquake is forthcoming.

The methods employed were using R programming language to perform wavelet analysis on GPS time series data, with a 30-day moving average filter applied, to find a repeating pattern in the GPS data along with its periodicity. We have specifically applied the Kolmogorov-Zurbenko filter (KZ) as a moving average. The research aimed to use the pattern along with the periodicity to determine if a deviation from the pattern could be used as an early indicator for an earthquake.

In the GPS data, depending on the direction of the GPS data, there were periods found of approximately 90, 120, 180, or 360 days, which most likely represent seasonal changes. Although there was a pattern found along with periods correlating with that pattern, there was no found correlation between this pattern and any deviations from it resulting in earthquakes. Additional methods and analysis of the data are needed in order to try to correlate these patterns to precursors for earthquakes.

Method/Materials:

- Nevada Geodetic Laboratory
 - GPS data from GPS stations in Japan
 - Up, North, East data
- RStudio (Version 2024.04.2+764)
 - Extrapolated data from NWIS
 - Fill in missing data (gaps in NWIS graphs)
 - Packages used: data.table, imputeTS, etc.
 - Moving average (kza)
 - Spectral analysis (periodograms)
 - Wavelet analysis

$$Y_t = \frac{1}{m} \sum_{j=-k}^k (X_{t+j})$$

Results:

- Identified periods in GPS data of approximately 90, 120, 180, and 360 days, likely linked to seasonal variations.
- Applied wavelet analysis and the Kolmogorov-Zurbenko (KZ) filter to detect patterns in GPS time series data.
- Observed recurring patterns in GPS data but no direct correlation between these patterns and earthquake events.
- Seasonal cycles were dominant in the analyzed GPS data, yet did not indicate impending earthquakes.

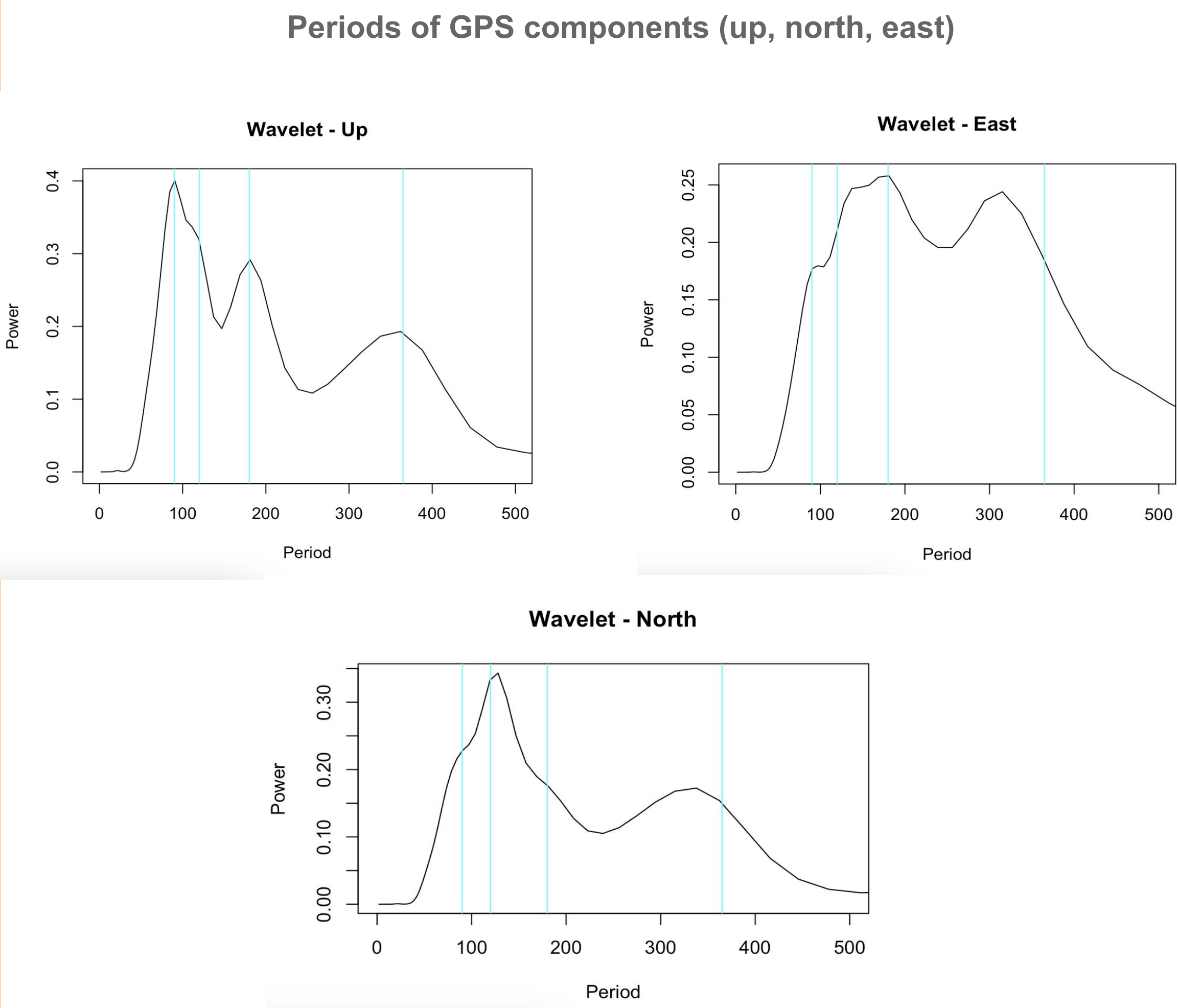


Figure 3: Wavelet analysis of 30-day KZ-average of coordinates of GPS station in Japan. Blue lines map 90, 120, 180, and 360 days accordingly.

Background Information:

- Japan is considered one of the most earthquake prone countries in the world, with the most infamous recent earthquake being the 2011 Tōhoku earthquake. (Which country has the most earthquakes?)
- The 2011 Tōhoku earthquake caused over 15000 deaths and over 2500 people reported missing. The earthquake caused an estimated \$220 billion dollars in damages. (NOAA)
- Due to the lack of knowledge of a clear pattern that corresponds with earthquakes, they are extremely difficult to predict. Furthermore, the build up to earthquakes - the colliding of tectonic plates, causing energy buildup - happens over long periods of time, which causes earthquakes to be even harder to predict. (What is an earthquake and what causes them to happen?)
- Earthquakes happen when the Earth's tectonic plates, which are constantly slowly moving, build up stress at their boundaries, and then suddenly release it, which causes the ground to shake. (What is an earthquake and what causes them to happen?)

Location

- The GPS data used in this analysis was in Erimo, Horozumi District, Hokkaido, Japan. The coordinates of the GPS station are 42° 00'21.6"N 143°09'25.2"E

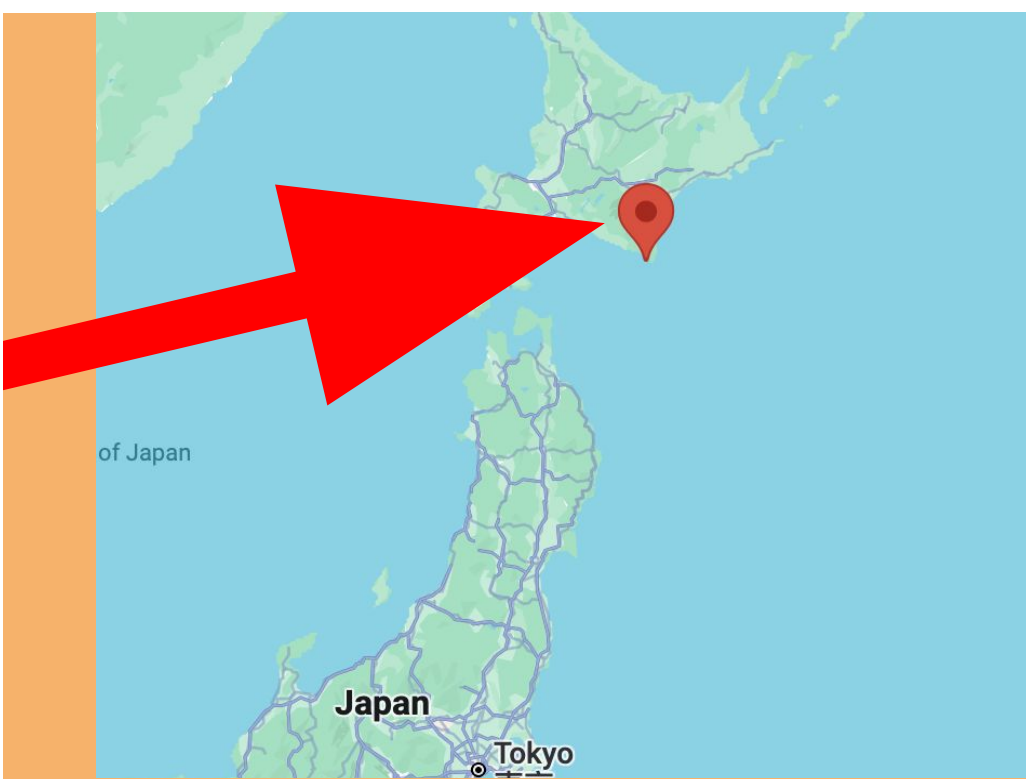


Figure 1: Location of GPS station used (Google Maps) in Japan.

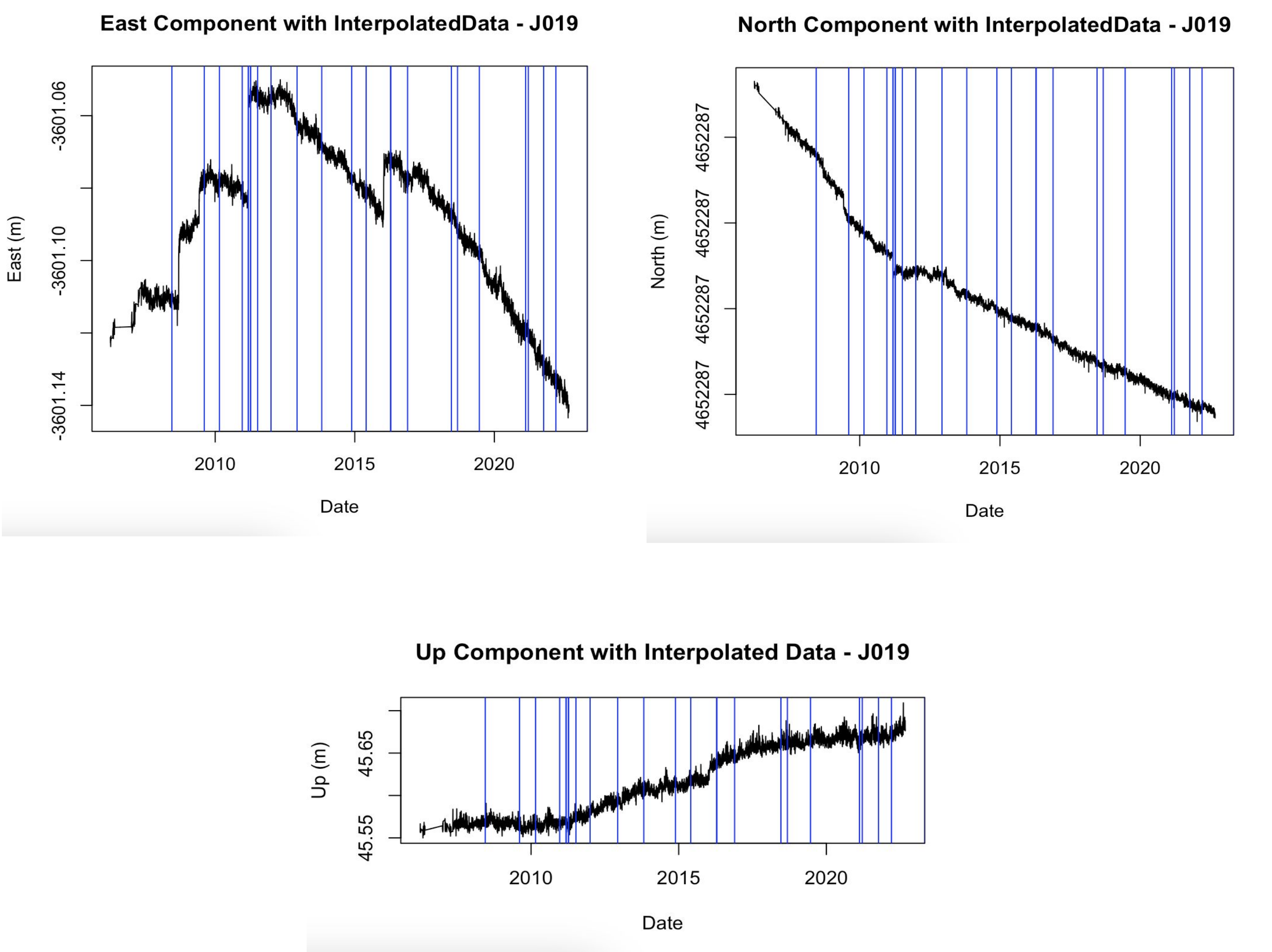


Figure 2: Raw GPS data showing East, North, and Up data accordingly. Blue lines represent major earthquakes (7.0 or higher magnitude)

Discussion/Conclusion:

- While clear periodic patterns were identified in the GPS data, no deviations from these patterns were found to predict earthquakes.
- Seasonal trends in GPS movement may be influencing the detected periodicity, but further research is needed to confirm any predictive value.
- The current analysis highlights the need for additional methods and data refinement to improve the detection of earthquake precursors using GPS time series.
- Future research should focus on refining the signal filtering process and exploring other data sources to enhance predictive capabilities.

Acknowledgements:

We would like to thank Dr. Antonios E. Marsellos and the Geology, Environment and Sustainability department at Hofstra University and HUSSRP for providing us with the necessary tools to conduct this research.

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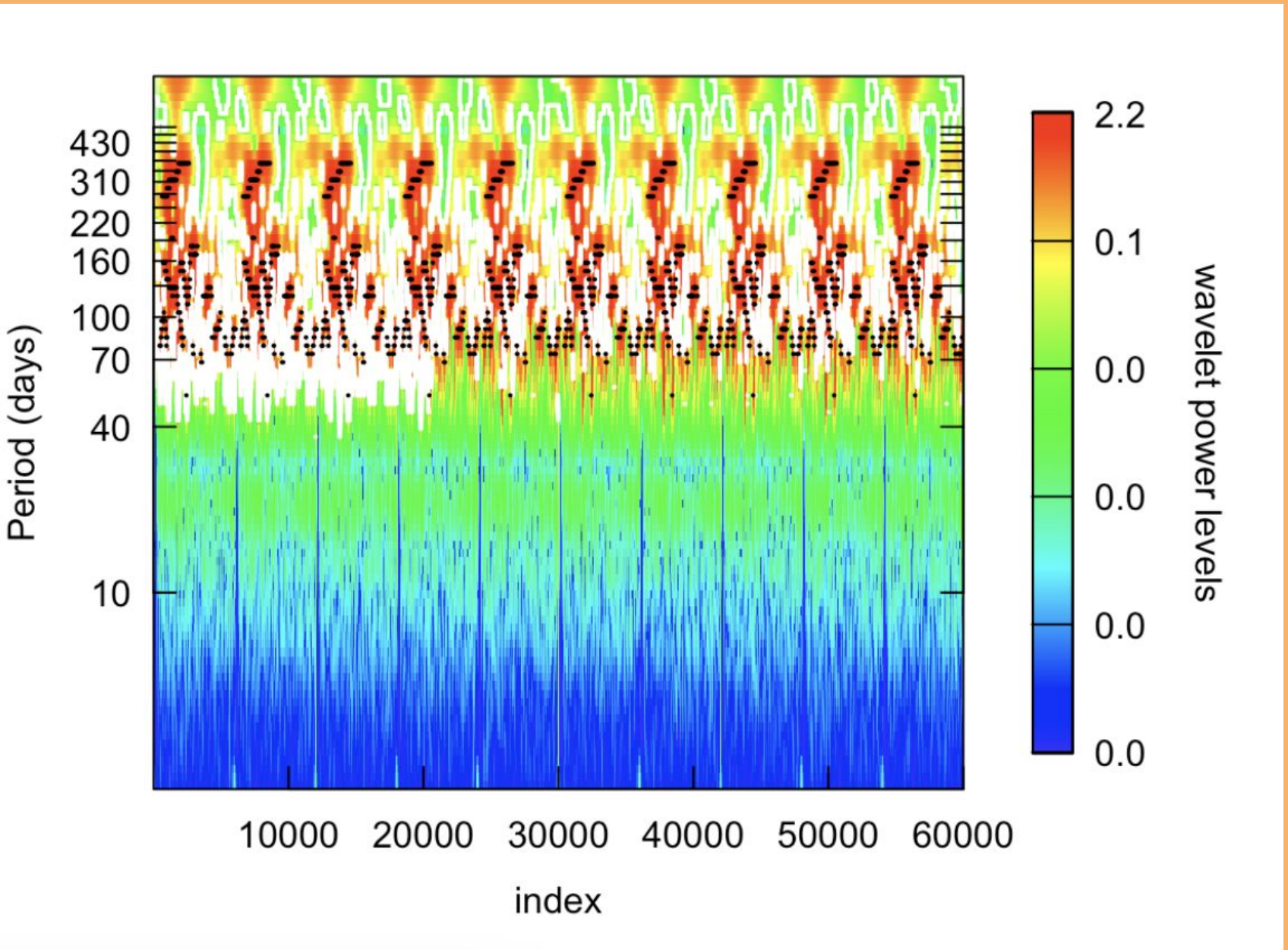


Figure 4: Wavelet analysis graph of the North component